
Advanced Computer Architectures

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Name
Last Name
ID Number

Problem 1 (20%)	
Problem 2 (15%)	
Problem 3 (15%)	
Question 1 (15%)	
Question 2 (20%)	
Question 3 (15%)	
Total (100%)	

NOTE: students have 90' to complete the exam.
To pass the exam, a minimum score of 25% is requested by each
“section” (section 1: problems, section 2: questions).

Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	lw \$1, 36(\$fp)	1	2	5	6
I2	lw \$3, 8(\$fp)	2	3	6	7
I3	lw \$4, 64(\$fp)	3	4	7	8
I4	lw \$2, 20(\$fp)	4	5	8	9
I5	sub \$3, \$2, \$3	5	10	11	12
I6	multdd \$f0, \$1, \$4	6	9	19	20
I7	add \$f1, \$f1, \$f0	7	21	25	26
I8	sd \$f1, \$3(fp)	10	27	30	31

- A. List all the possible conflicts in the code.
- B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?
- C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

Answer 1.A

Answer 1.B

Answer 1.C

A possible solution (NOT THE UNIQUE) is

Scoreboard with the following FUs: 4 LDU (3 cc latency), 1 ALU (1cc latency), 1 MUL (10 cc latency), 1 FPU (4 cc latency)

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	lw \$1, 36(\$fp)	1	2	5	6
I2	lw \$3, 8(\$fp)	2	3	6	7
I3	lw \$4, 64(\$fp)	3	4	7	8
I4	lw \$2, 20(\$fp)	4	5	8	9
I5	sub \$3, \$2, \$3	8	10	11	12
I6	multdd \$f0, \$1, \$4	9	10	20	21
I7	addd \$f1, \$f1, \$f0	10	22	26	27
I8	sd \$f1, \$3(fp)	11	28	31	32

Problem 2

In this problem, you will port code to a simple **4-issue VLIW machine**, and schedule it to improve performance.

Details about the 4-issue VLIW machine with 4 fully pipelined functional units:

- Integer ALU with **1** cycle latency to next Integer/FP and **2** cycle latency to next Branch
- 2 x Memory Unit with **2** cycle latency
- Floating Point Unit with 5 cycle latency for multiplication and 2 for other float operations
- Branch completed with **1 cycle delay slot** (branch solved in ID stage)

Assembly Code:

```
CONV: lw $1, 36($fp)
      lw $3, 8($fp)
      lw $4, 64($fp)
      lw $2, 20($fp)
      sub $3, $2, $3
      multdd $f0, $1, $4
      addd $f1, $f1, $f0
      sd $f1, $3(fp)
      bnez $3, CONV
```

P2.A Considering **one iteration** of the loop, **schedule** the assembly code for the 4-issue VLIW machine in the following table by using the as soon as possible (ASAP) **scheduling**, but neither using software pipelining nor loop unrolling **nor modifying loop indexes**. Please do not need to write in NOPs (can leave blank).

P2.B Answer the following performance questions

How long is the Critical Path?

What performance did you achieve in FP ops per cycle?

Answer 2.A

	ALU 1 cc	MEM 2 cc	MEM 2 cc	FPU */+ 5/2cc
C1		lw \$1, 36(\$fp)	lw \$3, 8(\$fp)	
C2		lw \$4, 64(\$fp)	lw \$2, 20(\$fp)	
C3				
C4	sub \$3, \$2, \$3			multdd \$f0, \$1, \$4
C5				
C6				
C7				
C8				
C9				addd \$f1, \$f1, \$f0
C10				
C11	bnez \$3, CONV		sd \$f1, \$3(fp)	
C12	(br delay slot)			
C13				
C14				
C15				

Please note that the bnez instruction has a control dependency.

Answer 2.B

How long is the Critical Path?

What performance did you achieve in FP ops per cycle?

Problem 3

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code (\$2 = 100, \$3 = 1000, \$0 = 1)

```
OUTER:    lw $1, 64($fp)
           lw $4, 20($fp)
           mult $3, $3, $2
INNER:    lw $1, 36($fp)
           div $3, $3, 10
           multdd $f0, $1, $4
           addd $f1, $f1, $f0
           sd $f1, $3(fp)
           bne $3, $0, INNER
           div $2, $2, 10
           bne $2, $0, OUTER
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

The INNER is executed @T0 5 times, while the OUTER 2 times. The second OUTER iteration INNER executes once only as \$3 is influenced by the OUTER behavior on \$2. Therefore, the worst case of the 2BHT (NTw_outer, NTs_inner with 6 mispredictions 2_inner+4_outer) will not be better than the best 1BHT case (T_outer, T_inner 2 mispredictions, 1_inner+1_outer). This is ok as 2BHT is better than 1BHT for two real nested loops.